

# Scalable Probit Share Calibration

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## Abstract

Probit models are a key primitive in economics, choice modeling, and many specialist areas of the humanities and natural sciences. Yet it would be a stretch to say they are the dominant paradigm and, to the contrary, each field has stumbled into the same technical difficulties when attempting to calibrate probit models at scale—almost invariably resorting to the more limited logit cousin. The well-cited probit methods, including those most commonly reached for in the R programming ecosystem for example, do not deliver repeated high-precision inversion with many alternatives. Monte Carlo and Quasi-Monte Carlo approaches have obvious drawbacks and orthant probability estimators that are smooth price one alternative at a time. In this paper we exhibit a new algorithm for calibrating utilities in a fixed factor probit model that, in contrast to prior art, is genuinely pragmatic at scale. The computed shares carry log-share errors of a few parts in a hundred thousand, hundred-to-one shares included. For problems with many variables, the performance improvements are very significant.

**Keywords:** multinomial probit; discrete choice; numerical quadrature; Jacobian–vector product; share inversion; factor model

## 1 Introduction

Probit models find their most obvious application in microeconomics, choice, and contest theory. Thurstone’s comparative judgment, the ordered and binary probits of applied microeconometrics, and the Lazear–Rosen tournament (Lazear and Rosen, 1981) are all the same Gaussian latent-performance model. Machine learning runs on the same family: a softmax output layer is its Luce–Gumbel member, and Thurstone–Mosteller pairwise comparison (Mosteller, 1951), the probit member, underlies rating systems and preference learning.

Races take many forms at all levels of spatial and temporal resolution. Predicting item selection and user clicks drives digital commerce, just as predicting market share

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is fundamental to company analysis many levels higher. Nature runs races constantly between competing chemical pathways, weak links in structures, molecular configurations and neural firing pathways.

Many of these races involve thousands of alternatives, as with competing nucleation sites in crystallization, crack-initiation locations, mutations, microbial lineages and transport channels. Any one of a thousand grain boundaries at which fracture, diffusion, segregation or corrosion can initiate is a competitor in a race. Other examples include precipitation nuclei during alloy aging, hydrogen-trapping sites in embrittlement, electrochemical reaction sites on an electrode, dendrite formation in a battery, charge trapping in semiconductors and breakdown paths through a dielectric.

Whether we consider fluid dynamics, atmospheric physics or machine learning, countless choices are made by man, nature or machine. Across all the world’s applications of large language models, perhaps a billion token sequences are chosen a second. Those microscopic battles are juxtaposed against the race for AI model supremacy.

From atoms to ecosystems, nature repeatedly poses the same computational problem: out of thousands, millions, or vastly more correlated alternatives—candidate crystal-growth directions, microscopic slip systems, adsorption geometries, pore-scale flow routes, turbulent modes, lightning leaders, fault asperities, phase-transition seeds, receptor-binding configurations, developmental cell fates, or ecological colonization opportunities—which one wins first?

Thus it is not only horses that race! However horse races are worthy of mention, and not only as regards attribution of a core trick used herein. For at least half a century those betting on them have been confronted by the nuances of choice and permutation probabilities in exacta, quinella, trifecta and place betting. In that setting there has been a strong Darwinian selection that weeds out those anchoring to a simple renormalization of probability—this does not model second place probabilities well at all if the favorite does not win.

At the racetrack both nature and the market make choices—the latter in risk-neutral probability. We don’t always think about markets in that manner but perhaps we should. Trillions of dollars are invested passively in capital weighted indexes in a subtle, unwitting nod to that same simple renormalization deemed unworthy for the racetrack.

The Capital Asset Pricing Model is internally coherent only for the universe of all assets, incidentally, and the argument that justifies it does not extend to sub-portfolios. In fact, the use of capital weighted indexes for investment or return comparison is fundamentally inconsistent in this respect—so in that sense it is in many ways analogous to the same unthinking renormalization that tends to occur almost everywhere in machine learning (including, of course, the very architecture of artificial neurons).

Choice models cannot be distinguished on a fixed set, assuming calibration is possible. However when logit, probit or other models are calibrated to observed shares<sup>1</sup> the recovered utilities can serve as inputs to questions the data do not answer directly: elasticities,

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<sup>1</sup>We call the choice probabilities  $p_i$  *shares* throughout. The racing origin (Cotton, 2021) called them *state prices*, borrowing prediction-market and finance usage. In the probit choice setting *shares*, as in market shares, is the more natural term, and *probability* or *implied probability* would blur them against the many other probabilities in play.

welfare, pricing and the effect of adding or removing an alternative and combinatorial probabilities. It is at this point that they make different, falsifiable predictions.

Although it is beyond the scope, we would suggest that Gaussian races are a priori more reasonable than Gumbel races for many if not most applications (the latter being equivalent to logit). However, judged by popularity the real workhorse of econometrics remains the logit family: multinomial logit, nested logit, and above all random-coefficients logit in the BLP tradition (Berry et al., 1995; Train, 2009; Conlon and Gortmaker, 2020). Logit models are universal for one reason. They can be computed.

Choice modeling did not start there, as an historical aside. It started with Thurstone’s 1927 model (Thurstone, 1927) of jointly Gaussian utilities with unrestricted correlation, brought into econometrics as the conditional (multinomial) probit model by Hausman and Wise (1978) and known since as MNP. The same object has re-emerged in machine learning: conditional on an input, heteroscedastic image classifiers place exactly this low-rank-plus-diagonal Gaussian race on their final layer and take the argmax, approximating the winner probabilities by Monte Carlo over a softmax relaxation (Collier et al., 2021).

Probit probabilities are  $(N - 1)$ -dimensional Gaussian integrals with no closed form. A well-cited approach is the Geweke–Hajivassiliou–Keane (GHK) simulator (Geweke et al., 1994; Hajivassiliou and McFadden, 1998; Train, 2009), the econometric branch of the Genz separation-of-variables family (Genz, 1992) whose RQMC form (Genz–Bretz) is the default in computational statistics. This method treats the probability that alternative  $i$  beats the rest as an orthant probability of the difference utilities, and estimates it by sampling truncated normals sequentially along a Cholesky factor. It is elegant but requires a separate simulation per alternative, and suffers  $R^{-1/2}$  error in the number of draws—not usually an issue for small menus but problematic for larger problems.

Consider the removal of an item from a list whose probabilities have been assigned, and the reflexive urge to apply simple renormalization. Recomputing new probabilities by assuming a Gaussian race is understandably rare (especially if that race involves a thousand competing choices of token, crystal lattice point, or segment of the sky). We might say that the field kept the computable models and gave up the model it started from. Simple linear renormalization of probability is one of the great unconscious acts, so natural we rarely stop to think about it, and built into the architecture of just about every neural network.

The concession is not free. Logit-family substitution is structurally restricted. One measure of the cost comes from an illustrative, deliberately misspecified synthetic design with oracle factor loadings (Section 6). Across twenty such designs, with every marginal variance held at one, we delete an alternative and ask where its demand goes. In the largest-deletion stratum plain logit misallocates a median 25% of the released demand, against 1% for factor probit (experiment 29), the closest of the four candidate structures.

This paper’s point is that probit’s computational barrier is not real for an important class of covariance structures, the low-rank-plus-diagonal factor model  $\Sigma = VV^\top + \text{diag}(D)$ . The goal is *calibration*: given observed shares and a fixed, known factor structure  $(V, D)$ , recover the utilities at which the probit model reproduces them.

We solve this fixed factor probit share calibration problem at  $N = 1000$  in three seconds. The numerical feat is the probit analogue of a far more popular logit inversion.

For example demand estimation runs on the logit side every day (Berry et al., 1995; Conlon and Gortmaker, 2020).<sup>2</sup>

Thus this paper counters the probit tractability objection. The computational reluctance that handed the popular victory to the logit family, almost universally, does not apply to factor-structured probit. The inverse problem is also globally well-posed (Proposition 2: shares determine utilities uniquely up to location).

Three ingredients make this possible:

1. *A fast forward map* (utilities  $\rightarrow$  shares): all  $N$  probabilities at once in  $O(QNL)$  for  $Q$  factor nodes and  $L$  lattice points. Calibration needs many forward evaluations; this is what makes them affordable.
2. *Resampling-free derivatives*: Jacobian–vector products in  $O(QNL)$ . In max-wins coordinates the Jacobian is a weighted graph Laplacian, so reduced systems are symmetric positive definite. This supports implicit differentiation and Newton–Krylov solvers. The calibration reported below uses a cheaper damped Jacobi iteration.
3. *A single shared construction*: conditionally on the  $k$  factors the alternatives are independent, and a lattice survival-product identity computes every alternative’s integral from one *shared survival field*, built once and reused for shares, slopes, and all  $N$  single-removal counterfactuals.

This one-pass identity comes from a mildly obscure but fitting literature: inferring ability from win probabilities in multi-entrant races (Cotton, 2021). The central trick this paper leans heavily on was described there as a multiplicity calculus: a characterization of tie density on a lattice and a set of algebraic operations for combining races and removing a runner. That is perhaps why it has not been assembled for probit before, where the focus has been on more classical methods for numerical computation.

## 2 The algorithm

The algorithm has one upfront limitation. It needs the factor structure. That structure is, as it happens, the original econometric model. Hausman and Wise (1978) induced correlation through random taste coefficients, giving a covariance that is exactly low-rank plus diagonal with observed attributes as loadings. Freeing the loadings is the factor-analytic MNP that McFadden (1984) proposed precisely to reduce the dimension of the integral.

**Model.** Throughout,

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \varepsilon_i, \quad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0, 1) \text{ i.i.d.}, \quad D_i > 0, \quad (1)$$

i.e.  $\Sigma = VV^\top + \text{diag}(D)$ ; the chosen alternative is the argmax of  $U$ . The pair  $(V, D)$  is supplied and held fixed. Calibration solves for  $\mu$  alone.

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<sup>2</sup>Estimating  $(V, D)$  is the outer problem, however, discussed in Section 7 but not claimed.

The algorithm rests on one fact. Conditional on the factors, the alternatives are independent. Write  $g_i(\cdot | f)$  and  $F_i(\cdot | f)$  for the conditional density and CDF of  $U_i$  given  $f$ . Alternative  $i$  wins when it beats every rival, so

$$p_i = \mathbb{E}_f \left[ \int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx \right], \quad \sum_i p_i = 1, \quad (2)$$

the shares summing to one because ties are null for continuous laws. The  $N$  products need not be formed separately. Compute  $\log F_{\text{field}} = \sum_j \log F_j$  once. Each product over  $j \neq i$  is then recovered by subtracting the own term.

The forward pass proceeds as follows. Lay down one common grid of  $L$  points, wide enough to contain every runner's distribution at every retained factor node. At each factor node  $f_q$ , shift each runner's location by  $v_i^\top f_q$  and evaluate each runner's log-survival curve on the grid. This is the log-probability that runner  $i$  has not yet finished by position  $x$ . Sum these  $N$  curves to obtain the *shared survival field*, the log-probability that nobody has finished by  $x$ . To price runner  $i$ , subtract its own log-survival back out of the field, multiply by its density, and integrate along the grid. This is the probability that  $i$  finishes first at this factor node, with everyone else behind.<sup>3</sup> Average over factor nodes with the quadrature weights, and normalize. Everything is done in the log domain so that deep-tail survivals never underflow. Algorithm 1 states this precisely.

A lattice of a few hundred points and a pruned product Gauss–Hermite rule suffice in practice. The exact resolution settings, and the evidence that they are converged, are collected with the numerical results in Section 4.

Algorithm 1 works in reflected min-wins coordinates  $a = -\mu$  and keeps the loadings' sign. The reflection is valid because the Gaussian law is unchanged under  $(f, \varepsilon) \mapsto (-f, -\varepsilon)$ .

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**Algorithm 1** All-share forward pass (min-wins; reflected from argmax by  $a = -\mu$ )

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**Require:** abilities  $a$ , loadings  $V$ , idiosyncratic variances  $D$ , factor nodes  $\{f_q, w_q\}_{q=1}^Q$ , lattice size  $L$

- 1:  $m_{iq} \leftarrow a_i + v_i^\top f_q$  for all  $i, q$
  - 2:  $[x_1, x_L] \leftarrow [\min_{i,q} m_{iq} - 8 \max_i \sqrt{D_i}, \max_{i,q} m_{iq} + 8 \max_i \sqrt{D_i}]$ , uniform spacing  $\Delta x$
  - 3: **for**  $q = 1, \dots, Q$  **do**
  - 4:    $\log S_i(x_\ell) \leftarrow \log \Phi(-(x_\ell - m_{iq})/\sqrt{D_i})$  for all  $i, \ell$   $\triangleright$  never  $1 - \Phi$
  - 5:    $\log S_{\text{field}}(x_\ell) \leftarrow \sum_j \log S_j(x_\ell)$   $\triangleright$  one shared field
  - 6:    $\tilde{p}_i += w_q \sum_\ell g_i(x_\ell) \exp\{\log S_{\text{field}}(x_\ell) - \log S_i(x_\ell)\} \Delta x$   $\triangleright$  divide one out
  - 7: **end for**
  - 8: **return**  $p = \tilde{p}/(\mathbf{1}^\top \tilde{p})$  and the diagnostic  $|1 - \mathbf{1}^\top \tilde{p}|$
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Per factor node, the conditional locations cost  $O(Nk)$ , the field log-product  $O(NL)$ , and each alternative's integral  $O(L)$  after subtraction, so one complete forward pass is

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<sup>3</sup>This divide-one-out step is the “removing a runner” operation of the multiplicity calculus in Cotton (2021), applied here once per alternative against a field built only once.

$O(QN(k + L))$ , which is  $O(QNL)$  for  $k \ll L$ . The same cached field also yields the full single-removal ensemble  $\{P(j \text{ chosen} \mid i \text{ removed})\}_{i,j}$  in  $O(QN^2L)$  arithmetic.

**Inversion.** Given target shares  $\hat{p}$  and  $(V, D)$ , the inverse transform finds mean-zero  $\mu$  with  $p(\mu) = \hat{p}$ .

Warm-start from a high-accuracy inversion of the matched *independent* race with matched total variances, which this same machinery provides using a single zero factor node. Then repeat the following. Run the forward pass at the current utilities. Read off each runner’s own-slope from the same pass. Take a coordinatewise Newton step on log shares. Cap and damp the step. Re-center the utilities to mean zero. Algorithm 2 states the loop. Its convergence is analyzed in Section 3 (Proposition 4).

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**Algorithm 2** Calibration (damped Jacobi quasi-Newton)

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**Require:** strictly positive target shares  $\hat{p}$  (normalized),  $V$ ,  $D$ , factor rule, tolerance  $10^{-6}$ , cap 50 iterations

- 1:  $a \leftarrow$  high-accuracy independent-race inversion using marginal variances  $D_i + \|v_i\|^2$  (this same algorithm with a single zero factor node)  $\triangleright$  warm start
- 2:  $\text{damp} \leftarrow 1$
- 3: **repeat**
- 4:   run Algorithm 1 at  $a$ , collecting  $\tilde{p}$  and the analytic own-location slopes  $\partial_i = \partial \tilde{p}_i / \partial a_i$  from the same pass
- 5:    $r_i \leftarrow \log(\tilde{p}_i / \mathbf{1}^\top \tilde{p}) - \log \hat{p}_i$ ; residual  $\leftarrow \max\{|r_i| : \hat{p}_i > 10^{-4}/N\}$
- 6:   halve the damping (floor 0.25) if the residual grew by more than 20%
- 7:    $a_i \leftarrow a_i - \text{damp} \cdot r_i / (\partial_i / \tilde{p}_i)$ , steps capped at  $\sqrt{D_i + \|v_i\|^2}$
- 8:    $a \leftarrow a - \bar{a}$   $\triangleright$  project to mean zero every update
- 9: **until** residual  $< 10^{-6}$

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The own-location slopes are the diagonal terms that Proposition 3 later makes explicit. The slopes in Algorithm 2 ignore the moving envelope and the output normalization.<sup>4</sup>

The tolerance is tail-aware. Alternatives with target shares below  $10^{-4}/N$  are matched best-effort. By Proposition 2 they remain uniquely determined, but they are statistically weakly resolved. This is also a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear.

Just as a horse at infinite odds sits outside the race, zero shares lie outside the proposition. The boundary targets correspond to utilities diverging to  $-\infty$ , so real assortment data requires pseudocounts or regularization, not merely the implementation floor. This adjustment must be left to the designer of the application. For example, depending on the data at hand, extrapolation to approximate shares could utilize near-misses. Alternatively, the user can simply not lean on those inferred probabilities that are very small.

Proposition 4 establishes local linear convergence of the exact continuum iteration. We do not establish global convergence of the capped, adaptive-grid implementation unfor-

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<sup>4</sup>They are Jacobi quasi-Newton preconditioners, not the exact diagonal of the adaptive map’s Jacobian.

tunately. We can only say that in practice it converges in low double digits of iterations, 6–31 across the stress suite of Section 4.

**Tail-stable implementation.** Tail stability requires the log domain throughout. We evaluate  $\log \Phi$  directly (forming  $1 - \Phi(z)$  by literal subtraction rounds to zero near  $z \approx 8.3$  through cancellation), use the analytic  $\log g$ , and form hazards as  $\exp(\log g - \log \Phi)$ . The Gaussian inverse Mills ratio<sup>5</sup> is unbounded but grows only linearly in the extreme tail, rather than exponentially, which is the numerically important property.

Flooring the density before taking its logarithm while the exact log-survival stays finite produces spurious  $e^{+92}$  hazards on problems as mild as  $N = 8$  with heterogeneous  $D$ . That configuration is a regression test. In the forward map the same discipline lets deep-tail shares resolve rather than floor to zero.

**Scope.** The method needs the factor form. Where  $\Sigma$  has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact  $\Sigma$  remains the tool (Section 6). The claim of scalability is in the number of alternatives at fixed modest rank. The quadrature size  $Q$  grows rapidly with the rank  $k$  (Section 4). The decisive advantages are precision, reproducibility, resampling-free derivatives, and the calibration machinery.

### 3 What the algorithm achieves

The algorithm computes a particular map from utilities to shares. This section records what that map is like: its identification and derivative structure, the well-posedness of its inverse, the graph that organizes both, and several interpretations that connect it to older mathematics.

**Proposition 1** (Identification and differentiability).  *$p(\mu + c\mathbf{1}) = p(\mu)$ , so  $J = \partial p / \partial \mu$  has  $\mathbf{1}$  in its null space and  $\mu$  is identified only up to location. With  $B$  an  $N \times (N - 1)$  basis of  $\mathbf{1}^\perp$  and  $\mu = B\eta$ , share inversion and implicit differentiation are taken in reduced coordinates:  $d\eta/d\theta = -(B^\top JB)^{-1}B^\top \partial p / \partial \theta$ , provided  $B^\top JB$  is nonsingular.*

*Proof.* For the invariance, write  $U_i = \mu_i + \xi_i$  with  $\xi_i = v_i^\top f + \sqrt{D_i} \varepsilon_i$ . For every realization of the shocks,  $\arg \max_i (\mu_i + c + \xi_i) = \arg \max_i (\mu_i + \xi_i)$ , since adding  $c$  to every coordinate changes no pairwise comparison. The choice events therefore coincide realization by realization, so  $p(\mu + c\mathbf{1}) = p(\mu)$  for all  $c$ . Differentiating this identity in  $c$  at  $c = 0$  gives  $J\mathbf{1} = 0$ : the ones vector lies in the null space of  $J$ , and two utility vectors differing by a common shift produce identical shares, so  $\mu$  is identified at most up to location.

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<sup>5</sup>The inverse Mills ratio is  $\phi(z)/\Phi(z)$ , the ratio of the standard normal density to its cumulative distribution function. It is the conditional hazard in this setting, and behaves like  $|z|$  as  $z \rightarrow -\infty$  because  $\Phi(z) \approx \phi(z)/|z|$  in the deep tail.

For the reduced formula, suppose  $\eta(\theta)$  satisfies the calibration identity  $p(B\eta(\theta), \theta) = \hat{p}$  on an open set of  $\theta$ . Total differentiation gives the  $N$ -equation system

$$JB \frac{d\eta}{d\theta} + \frac{\partial p}{\partial \theta} = 0.$$

Only  $N - 1$  of these equations are independent, but none are lost by projecting: shares sum to one identically in both arguments, so differentiating  $\mathbf{1}^\top p = 1$  gives  $\mathbf{1}^\top J = 0$  and  $\mathbf{1}^\top (\partial p / \partial \theta) = 0$ ; both terms already lie in  $\mathbf{1}^\perp$ , so the system is equivalent to its image under  $B^\top$ . That image is  $(B^\top JB) d\eta/d\theta = -B^\top \partial p / \partial \theta$ , and when  $B^\top JB$  is nonsingular (established for this model in Proposition 2) it inverts to the displayed formula, with local existence and smoothness of  $\theta \mapsto \eta(\theta)$  supplied by the implicit function theorem.  $\square$

**Proposition 2** (Shares determine utilities). *For model (1) with  $D_i > 0$  and fixed known  $(V, D)$ , in max-wins coordinates, define for  $i \neq j$*

$$w_{ij} = \mathbb{E}_f \int g_i(x | f) g_j(x | f) \prod_{\ell \neq i, j} F_\ell(x | f) dx > 0.$$

*Then  $J = \partial p / \partial \mu$  satisfies  $J_{ij} = -w_{ij}$  and  $J_{ii} = \sum_{j \neq i} w_{ij}$ , i.e.*

$$J = \sum_{i < j} w_{ij} (e_i - e_j)(e_i - e_j)^\top, \quad h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2.$$

*This is a weighted graph Laplacian with everywhere-positive weights. Furthermore  $J$  is symmetric, positive semidefinite, has  $\mathbf{1}$  as its only null direction, and  $B^\top JB \succ 0$  for every full-rank basis  $B$  of  $\mathbf{1}^\perp$ . Moreover  $p = \nabla G$  for the social-surplus function  $G(\mu) = \mathbb{E} \max_i \{\mu_i + \xi_i\}$ . This is the Williams–Daly–Zachary / McFadden identity (Williams, 1977; Daly and Zachary, 1978; McFadden, 1981). The restriction of  $p$  to  $\mathbf{1}^\perp$  is a smooth bijection onto the interior of the probability simplex. Every strictly positive target  $\hat{p}$  has exactly one mean-zero utility vector.*

*Sketch.* Off the diagonal  $\mu_j$  enters  $p_i$  only through  $F_j$ , giving  $J_{ij} = -w_{ij}$ ; translation invariance fixes the diagonal, so  $J$  is the stated weighted graph Laplacian, positive definite on  $\mathbf{1}^\perp$  because every weight is positive. The inverse is the unique minimizer on  $\mathbf{1}^\perp$  of the strictly convex, coercive potential  $H(\mu) = G(\mu) - \hat{p}^\top \mu$ , and the inverse-function theorem gives smoothness. Appendix A carries out each step.  $\square$

**Covariance is not fit.** Surjectivity cuts both ways. For any fixed admissible  $(V, D)$ , every interior full-menu share vector is fitted exactly by some utilities, so a perfect same-menu fit carries no evidence that the covariance is right. The covariance is disciplined only by restrictions across menus, markets, or covariates, or by micro-level data (Chang et al., 2026). Its substantive content lies in the substitution and counterfactual predictions measured in Section 6.



**The photo-finish graph.** Given the origins of the central numerical insight, the reader will forgive us for speaking about Proposition 2 in racetrack terminology. Here  $w_{ij}$  is the probability density of a photo-finish between  $i$  and  $j$ , a tie for the win with everyone else behind, and moving utility from  $j$  to  $i$  moves share along every photo-finish edge at exactly that rate. The alternatives form a *photo-finish graph*: complete, with strictly positive weights, because Gaussian utilities give every pair a strictly positive tie density at the winning frontier.

The proposition’s conclusions are then graph facts. Constants span the null space because the quadratic form  $\sum_{i < j} w_{ij} (h_i - h_j)^2$  sees only differences. Shares cannot identify the overall utility level, only gaps, which is economically obvious. The odds do not inform us of the class of the race.

Positive weights make the graph connected, so the constant vector is the *only* null direction, the gradient map is strictly monotone as an operator on  $\mathbf{1}^\perp$  (not coordinatewise), and shares determine utilities uniquely.

And symmetry plus positive definiteness make every reduced Newton system of the continuum map symmetric positive definite, so conjugate gradients apply with the  $O(QNL)$  Jacobian–vector product of Proposition 3 playing the role of a fast Laplacian multiply. Each linearization of the calibration solves a weighted graph-Laplacian system on the photo-finish graph, with weights that move with  $\mu$ .

The graph is complete and its  $O(N^2)$  weights are never materialized, so the benefit is matrix-free Krylov solving and spectral interpretation rather than the complexity guarantees of sparse-Laplacian solvers. The implemented map’s derivative differs from the continuum Laplacian by small normalization and grid terms, quantified under *the differentiated map* below.

At this point nothing is specifically Gaussian, just as nothing in Cotton (2021) requires horse performance to be Gaussian. Any conditionally independent additive random-utility race with everywhere-positive smooth densities has, under the corresponding integrability and differentiability conditions (finite social surplus, differentiation under the integral), a weighted-Laplacian Jacobian and an interior-simplex diffeomorphism modulo translation. Gaussian factor probit is the computationally important specialization.

**Relation to the multiplicity calculus.** The photo-finish weight is the differentiated form of the *multiplicity* that the original transform (Cotton, 2021) tracks in the forward direction. There the multiplicity is a scalar field on the lattice: the expected number of runners sharing the winning cell, needed to split dead-heat payoffs, since a  $k$ -way tie pays each winner  $1/k$ .

Multiplicity so defined is an expectation of a discrete count, an excess over one that is  $O(\Delta x)$  in the cell width and vanishes as the lattice refines. The weight  $w_{ij}$  is the continuum, pairwise density of the same event, a two-way tie between  $i$  and  $j$  with the rest behind, and it survives the continuum limit because  $-w_{ij} = \partial p_i / \partial \mu_j$  is a rate rather than a count. The two are one identity apart: the multiplicity’s excess over one, per unit

lattice spacing, is the total tie density

$$\sum_{i < j} w_{ij} = \frac{1}{2} \text{tr}(J),$$

which experiment 37 confirms—the measured excess-over-one divided by the cell width converges to  $\frac{1}{2} \text{tr}(J)$  as  $\Delta x \rightarrow 0$ . The forward pass counts ties to divide payoffs; calibration differentiates them into the graph-Laplacian weights that make it a network solve.

**The conjugate view.** Here we lean on social-surplus convex duality for discrete choice (Chiong et al., 2016; Norets and Takahashi, 2013). Our contribution is not the well-posedness of the algorithmic challenge but rather its solution.  $G$  is convex with  $p = \nabla G$ , so  $\Omega = G^*$  is a convex regularizer on the simplex and  $p(\mu) = \arg \max_{q \in \Delta} \{\mu^\top q - \Omega(q)\}$ . For iid Gumbel noise  $\Omega$  is Shannon negentropy and the maximizer is softmax. In general it is the generalized entropy of Fosgerau et al. (2020), with the perturbation–regularization duality developed for machine learning by Berthet et al. (2020). Factor probit carries a covariance-dependent, non-separable entropy that couples alternatives through the factor structure.

In this paper we give three algorithms, and they could be thought of as three faces of that dual pair. The transform evaluates  $\nabla G$ . Calibration evaluates the mirror map  $\mu = \nabla \Omega(p)$  on the mean-zero quotient. The Jacobian–vector product applies the Hessian  $\nabla^2 G$ . Each is built from the same  $O(QN(k + L))$  primitive. The forward map is one pass, each Jacobian–vector product is one pass, and each calibration iteration is one pass (total calibration cost multiplies by the iteration count, typically 10–15).

On the simplex,  $\nabla \Omega(p)$  is an equivalence class modulo constants. Its mean-zero representative is the calibrated utility. The bijection itself is essentially known: injectivity up to a common shift is the Hotz–Miller inversion (Hotz and Miller, 1993), surjectivity is Norets and Takahashi (2013), and Berry et al. (2013) derive demand invertibility from connectivity of the substitution structure, the same mechanism as the photo-finish graph. The appendix proof (Appendix A) keeps the Gaussian factor case self-contained and exhibits the weights the algorithm uses. Each identity in the proposition is also checked numerically in the repository: the  $w_{ij}$  representation against finite differences to  $6 \times 10^{-9}$ , symmetry to  $4 \times 10^{-12}$ , and positive definiteness of the reduced numerical Jacobian.

**Transport form.** There is another interpretation. On the contrast space, with  $b_i = e_i - \mathbf{1}/N$  the vertices of a regular simplex,  $\arg \max_i (x_i + \mu_i) = \arg \min_i \{\frac{1}{2} \|x - b_i\|^2 - \mu_i\}$ : the choice cells are the Laguerre cells of quadratic semi-discrete optimal transport from the contrast Gaussian to  $\nu_q = \sum_i q_i \delta_{b_i}$ , and Kantorovich duality identifies  $\Omega(q)$ , up to an additive constant, with  $\frac{1}{2} W_2^2$  between them. Calibration is the dual solve of a structured transport problem;  $w_{ij}$  is the Gaussian surface mass on the shared Laguerre facet (scaled by  $1/\sqrt{2}$ ), and a Newton step is an electrical solve on that network, for which Proposition 3 is the matrix-free multiply.<sup>6</sup>

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<sup>6</sup>Place the alternatives as the corners of a regular simplex, and let each Gaussian draw pick its nearest corner, where corner  $i$ ’s reach is enlarged or shrunk by its utility  $\mu_i$ . The choice regions are then the

The geometry above is standard semi-discrete optimal transport. The power-diagram formulation goes back to Aurenhammer et al. (1998). Lévy (2015) optimizes the weights through a convex objective. Kitagawa et al. (2019) analyze a damped Newton method whose Hessian is a weighted graph Laplacian with facet-integral weights, with global convergence under regularity and support conditions that an unbounded Gaussian source does not immediately satisfy. Taşkesen et al. (2023) connect discrete-choice smoothing and semi-discrete transport directly. What is new here is the oracle. The factor structure gives an  $O(QN(k + L))$  evaluation of all cell masses at once, and of Hessian–vector products without facet enumeration, when the source is a factor Gaussian and the sites are the simplex vertices.

At calibrated utilities the conjugate value has a closed form. Gaussian integration by parts gives  $G(\mu) = \mu^\top p + \text{tr}(\Sigma J)$ , so the regularizer is

$$\Omega(p) = -\text{tr}(\Sigma J) = -\sum_{i < j} w_{ij} s_{ij}^2, \quad s_{ij}^2 = D_i + D_j + \|v_i - v_j\|^2.$$

This brings us to a remarkable circuit interpretation. Read the graph as a network of wires:  $w_{ij}$  is the conductance of edge  $ij$ , and  $s_{ij}^2 = \text{Var}(U_i - U_j)$  is the noise in the  $i$ -versus- $j$  race. The entropy  $\Omega$  is minus the total conductance, each edge weighted by its noise. Only differences  $v_i - v_j$  and sums  $D_i + D_j$  appear, so adding a common part to the covariance leaves  $\Omega$  unchanged—the contrast-only property again.

We checked this in the code (see experiment 27 in particular). The binary closed form  $\Omega = -s \phi(\Phi^{-1}(q))$  to  $3 \times 10^{-13}$  and the full identity to  $3 \times 10^{-9}$  are both verified. That  $G^*$  is an assignment problem is previously known per Chiong et al. (2016) but new here is the simplex form and also the evaluation of the conjugate value at calibrated utilities from  $k$  Jacobian–vector products and the diagonal slopes. The gradient  $\nabla \Omega(p) = \mu$  is dearer. It requires the calibration solve, and Hessian actions of  $\Omega$  require reduced linear solves built from the same Jacobian–vector products.

**Proposition 3** (Jacobian–vector products in  $O(QNL)$ ). *With hazards  $\lambda_j = g_j/F_j$ ,  $\Lambda = \sum_j \lambda_j$ , and  $A = \sum_j h_j \lambda_j$  (all conditional on  $f$ , all functions of  $x$ ), the Jacobian of the exact max-wins map acts on a direction  $h$  as*

$$(Jh)_i = \mathbb{E}_f \int g_i \left( \prod_{j \neq i} F_j \right) (h_i \Lambda - A) dx,$$

*where the field sums  $\Lambda$  and  $A$  are formed once per node and lattice point, so all  $N$  components cost  $O(NL)$  per node. In effect, a direction  $h$  perturbs alternative  $i$ ’s share only through the gap between its own hazard weight  $h_i \Lambda$  and the field average  $A$ . Translation*

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cells of a power diagram (a weighted Voronoi tessellation), and a choice probability is the Gaussian mass that falls in a cell. Calibration—nudging the utilities until each cell captures its target share  $q_i$ —is the semi-discrete optimal-transport problem of moving the Gaussian mass onto the corners at least squared-distance cost, with the utilities acting as transport prices and  $\Omega$  as the transport cost. The weight  $w_{ij}$  is the Gaussian mass on the wall between cells  $i$  and  $j$ ; a Newton step pours the share residuals into the nodes as currents, treats those wall-masses as conductances, and solves for utility corrections as voltages, which Proposition 3 performs without forming the matrix.

invariance is visible ( $h = \mathbf{1}$  gives  $A = \Lambda$ , hence  $Jh = 0$ ), and the weighted-Laplacian form of Proposition 2 follows by expanding  $A$ . This enables Krylov solves of the reduced system in Proposition 1 without forming  $J$  densely at  $O(QN^2L)$ .

*Proof.* Write  $R_i = \prod_{j \neq i} F_j$ . For the own derivative,  $\partial_{\mu_i} g_i(x) = -g'_i(x)$  (a location shift), and integrating by parts,  $\int (-g'_i) R_i dx = \int g_i R'_i dx = \int g_i R_i \sum_{j \neq i} \lambda_j dx = \int g_i R_i (\Lambda - \lambda_i) dx$ . For  $j \neq i$ ,  $\partial_{\mu_j} F_j = -g_j$ , so the cross derivative is  $-\int g_i g_j \prod_{\ell \neq i, j} F_\ell dx = -\int g_i R_i \lambda_j dx$ . Summing against  $h$ ,  $(Jh)_i = \int g_i R_i [h_i (\Lambda - \lambda_i) - \sum_{j \neq i} h_j \lambda_j] dx = \int g_i R_i (h_i \Lambda - A) dx$ ; average over  $f$ .  $\square$

**The differentiated map.** Because the algorithms presented here use lattice computation, we will be clear about the distinction between the approximation and the continuous idealization (just as in Cotton (2021) we delineated the discrete horse race from the continuous).

Due to conventions three maps must be distinguished: the exact max-wins map  $p^{\max}(\mu)$ ; the reflected min-wins map  $p^{\min}(a)$  used by the implementation, related by  $a = -\mu$  so that  $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$ ; and the implemented approximation  $p_{Q,L} = \tilde{p}_{Q,L} / (\mathbf{1}^\top \tilde{p}_{Q,L})$ , whose derivative carries a normalization correction: with  $v = D\tilde{p}[h]$  and  $s = \mathbf{1}^\top \tilde{p}$ ,

$$Dp_{Q,L}[h] = \frac{v - p_{Q,L}(\mathbf{1}^\top v)}{s}.$$

The correction vanishes for the continuum map and is numerically negligible at the resolutions tested. A further distinction is that integration by parts is a continuum identity. The Laplacian formula of Proposition 3 is therefore the exact derivative of the *continuum* map, not of the finite rectangle sum.

The exact derivative of the frozen-grid sum replaces the own term by  $(x - m_i)/D_i$  (no integration by parts) and costs the same  $O(QNL)$ . That is why we have shipped it in the repository (`form="grid"`).

Measured at  $L = 1501$  (the derivative-comparison resolution) the two forms agree to  $1.7 \times 10^{-14}$ . At  $L = 51$  the continuum form is off by  $2.6 \times 10^{-5}$  while the grid form still matches finite differences. That is why we place lesser emphasis on the difference. (We make the passing remark that in practice the Laplacian form has proven to be an extremely accurate symmetric surrogate at the resolutions used). But again, the exact differentiation of the normalized *frozen-grid* map is available when needed. The returned adaptive map additionally moves its envelope with  $\mu$ , a tail effect measured below  $10^{-8}$  at  $L = 1501$ .

The implemented inversion rebuilds the lattice envelope from the current utilities at each iteration, holds it fixed within the iteration, and projects the utilities to mean zero after every update (Algorithm 2). The same graph controls the convergence of that loop.

**Proposition 4** (Local convergence of damped Jacobi). *For the exact continuum share map and a strictly positive target, the mean-zero damped Jacobi iteration underlying Algorithm 2 is locally linearly convergent for every fixed damping  $0 < \alpha < 1$ , and also at  $\alpha = 1$  for  $N \geq 3$ , with asymptotic local convergence factor  $\max_{j \geq 2} |1 - \alpha \lambda_j|$ , the spectral*

rate on the mean-zero quotient, where  $\lambda_j$  are the nonzero eigenvalues of the random-walk normalized photo-finish Laplacian  $\Delta^{-1}J$ ,  $\Delta = \text{diag}(J_{11}, \dots, J_{NN})$ .

*Proof.* With  $r(\mu) = \log p(\mu) - \log \hat{p}$ , the Jacobian at the solution is  $Dr = \text{diag}(p)^{-1}J$ , and the iteration’s exact Jacobi preconditioner is  $M = \text{diag}(J_{ii}/p_i)$ , so the linearized update on the mean-zero quotient is  $I - \alpha\Delta^{-1}J$ . By Proposition 2,  $J$  is a graph Laplacian with strictly positive weights, so  $\Delta^{-1}J$  is similar to the symmetric normalized Laplacian  $\Delta^{-1/2}J\Delta^{-1/2}$ , whose spectrum lies in  $[0, 2]$  with 0 simple (the common-shift direction, removed by the quotient). The photo-finish graph is complete, hence nonbipartite for  $N \geq 3$ , so  $\lambda_N < 2$  and  $\max_{j \geq 2} |1 - \alpha\lambda_j| < 1$  for  $0 < \alpha \leq 1$ . Step caps are inactive near the solution, and the mean-zero projection selects a quotient representative without altering the spectrum.  $\square$

Experiment 31 measures this at  $N = 12$ . The spectrum of  $\Delta^{-1}J$  is  $[0, 0.974, \dots, 1.379]$ , and the observed contraction of the exact iteration matches the predicted factor to two decimals (0.369 against 0.379 at  $\alpha = 1$ ; 0.316 against 0.318 at  $\alpha = 0.7$ ). Global convergence of the capped, adaptive-grid implementation remains empirical; slow convergence corresponds to a small normalized spectral gap  $\lambda_2$ , which the photo-finish graph makes interpretable. Tiny shares or nearly redundant alternatives weaken the graph’s connectivity.<sup>7</sup>

**Beyond Gaussian factors.** The factors need not be Gaussian because (2) requires only conditional independence and an integrable factor law. Other obvious candidates include Student, mixture, or empirical factor distributions. Also in keeping with Cotton (2021) the base densities may be arbitrary and alternative-specific, modulo the numerical comments we make throughout that might apply to adversarially designed mixtures of Dirac distributions and the like intended to test the limits of the lattice computation.<sup>8</sup>

## 4 Numerical evidence of accuracy

This section assesses the algorithm in isolation. Comparisons with other methods are deferred to Section 6. The evidence, in the order presented: correctness anchors on cases with closed forms; forward accuracy replicated across problem sizes against twin simulation references (experiment 16); resolution sweeps of the two quadrature knobs (experiment 17); a grid stress test under extreme variance heterogeneity (experiment 33); log-share accuracy against higher-resolution internal references (experiment 25); an error

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<sup>7</sup>In circuit terms, damped Jacobi preconditions each node by its own total conductance and ignores the wiring, so it is fast exactly when the network is well connected. The other correspondences are just as literal. Kirchhoff’s current law is the constraint that shares sum to one, and the irrelevance of a common voltage offset is the location non-identifiability.

<sup>8</sup>Another brief caution arising from sign conventions might be in order as well. For asymmetric laws one should be careful since the implementation works in the reflected (min-wins) race.  $-U$  has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

budget separating rigorous bounds from convergence evidence; calibration accuracy, cost, and scaling at  $N = 1000$  to  $10000$  (experiments 16, 19, and 21); derivative accuracy; and two by-products, estimation gradients (experiment 30) and the removal ensemble (experiment 18).

**Settings.** Unless otherwise stated, forward and calibration runs use  $L = 501$  lattice points, chosen from the measured lattice sweep (experiment 17 spans  $L = 101$  to  $6001$ , with self-convergence reaching machine precision by  $L \approx 200$ ). The derivative comparison below uses  $L = 1501$ , and higher-resolution reference calculations use  $L = 3001$  or  $5001$ . The lattice sits on the adaptive interval  $[\min_{i,q} m_i(f_q) - 8 \max_i \sqrt{D_i}, \max_{i,q} m_i(f_q) + 8 \max_i \sqrt{D_i}]$ , where  $q = 1, \dots, Q$  indexes the retained factor nodes. The rectangle rule uses  $\Delta x$  weights with no endpoint modification, differing from the trapezoidal rule only by negligible endpoint terms.

The main factor integration experiments use product Gauss–Hermite of order  $15/11/9$  for  $k = 2/3/4$ . The boundary experiment (experiment 14) uses order 15 per dimension for all  $k \leq 4$ , which after pruning weights below  $10^{-7}$  of the maximum leaves  $Q = 145$ ,  $1317$ , and  $10929$  nodes for  $k = 2, 3, 4$ , making the exponential growth of  $Q$  with rank explicit. Scrambled-Sobol nodes ( $2^{13}$  points, fixed seed) are used only for  $k > 4$ . They are reproducible by construction, though not deterministic in the operational sense. Survival floors sit at  $10^{-300}$  and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2). Statements about derivatives refer to derivatives of this approximation, with fidelity governed by  $Q$  and  $L$ .

Pruned factor weights are not renormalized explicitly. The final normalization conditions the quadrature on the retained node set, and the two contributions to the pre-normalization defect separate cleanly. The pruned factor weight is  $\delta_{\text{factor}} = 2.39 \times 10^{-8}$  at  $k = 2$  (analytic). The measured  $|1 - \mathbf{1}^\top \tilde{p}| = 2.4 \times 10^{-8}$  equals it, so lattice truncation contributes below that level.

**Correctness anchors.** On the binary case, where MNP has a closed form, the lattice transform agrees to  $3.3 \times 10^{-16}$  (an easy anchor, useful for code validation rather than as evidence about the hard regime).

At  $N = 5$  the lattice transform and a large-draw simulator both sit within the noise of a  $10^7$ -draw Monte Carlo reference, and the shipped package implementation agrees with the benchmark kernel to  $1.1 \times 10^{-5}$  (the package default uses its own coarser lattice and interval rule; at matched settings agreement is machine precision).

**Error metric.** We report two metrics. The first is the maximum absolute share error  $\max_i |\hat{p}_i - p_i|$  against a simulation reference. Absolute error is not always the decision-relevant scale, though. In betting, moving a 100-to-1 chance to 150-to-1 is an enormous change carried by only  $|\Delta p| \approx 0.003$ . So we also report the log-share error  $\max_i |\log p_i^{\text{lat}} - \log p_i^{\text{ref}}|$  over resolvable shares (which agrees with log-odds error to  $O(p)$  on small shares). The log-domain lattice is at its strongest on this second metric. Deep-tail shares resolve to genuine small positives, whereas a direct simulation reference resolves relative error  $\epsilon$  only

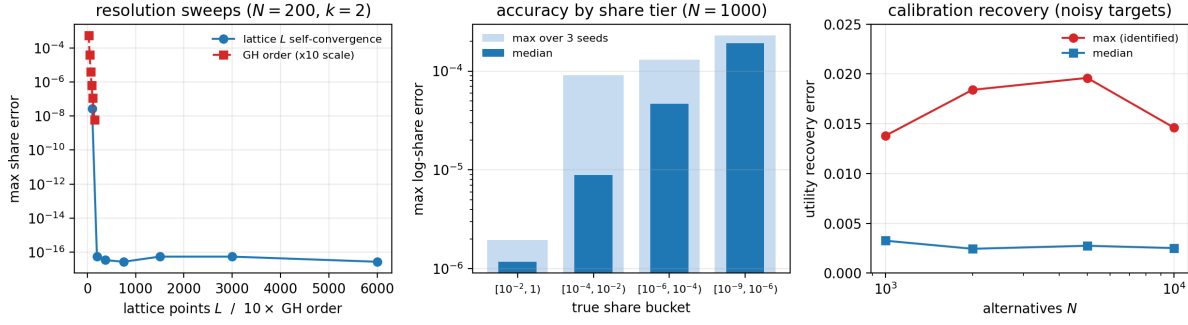


Figure 1: Central validation evidence, from the experiments. Left: lattice self-convergence in  $L$  and Gauss–Hermite factor sweep (experiment 17). Middle: log-share error against higher-resolution internal references by true-share bucket at  $N = 1000$ , three problems (experiment 25). Right: utility recovery against independent  $5 \times 10^6$ -draw noisy targets over  $N = 1000$  to 10000 (experiment 19); the plateau is target sampling noise, not solver error (experiment 21).

for shares above roughly  $1/(R\epsilon^2)$ . A maximum over  $N$  coordinates tends to grow with  $N$  at fixed per-coordinate accuracy, and the reference’s own maximum-coordinate noise behaves likewise. Mean-absolute and total-variation summaries are reported alongside (experiment 16). At  $N = 1000$  the lattice’s mean absolute error is  $5.6 \times 10^{-6}$  and total variation  $2.8 \times 10^{-3}$  against a fresh  $2 \times 10^6$ -draw reference.

**Replication.** Ten problems per size at  $N \leq 200$ , three at  $N = 1000$ , each scored against twin independent  $2 \times 10^6$ -draw references (experiment 16): worst-case lattice error  $3.5$ – $4.6 \times 10^{-4}$  at every size, at or below the references’ own replicate-to-replicate noise. The convergence evidence indicates accuracy far below this measurement floor (the resolution sweeps and the log-odds assessment below).

**Log-share assessment.** Against higher-resolution internal references (Gauss–Hermite  $60^2$  with  $L = 5001$  through  $N = 100$ ;  $40^2$  with  $L = 3001$  at  $N = 1000$ , three problems; experiment 25) the production settings’ log-share error is at most  $5 \times 10^{-5}$  on every share from  $10^{-9}$  to one at  $N \leq 100$ . At  $N = 1000$  it is at most  $2 \times 10^{-6}$  on shares above one percent,  $9 \times 10^{-5}$  down to  $10^{-4}$ , and  $2.3 \times 10^{-4}$  on the deepest tails (shares to  $10^{-9}$ ). A plain winner-frequency simulation could not resolve these relative tail errors at practical draw counts: it needs about  $1/(p\epsilon^2)$  draws for relative error  $\epsilon$  on a share  $p$ .

**Resolution sweeps.** Figure 1 collects the central accuracy evidence. Experiment 17 sweeps each resolution knob separately at  $N = 200$ ,  $k = 2$ . Holding the factor rule fixed, lattice self-convergence is  $2.7 \times 10^{-8}$  at  $L = 101$  and at machine precision from  $L \approx 200$ . The convergence observed over the tested range is near-exponential. For these analytic, rapidly decaying integrands the lattice error becomes negligible very quickly, leaving the  $L$ -error subdominant.

Holding  $L$  fixed, the Gauss–Hermite sweep decays from  $5.3 \times 10^{-4}$  (order 3) to  $5.9 \times 10^{-9}$  (order 15).

**Experimental errors.** Empirical convergence evidence is not a rigorous proof, and we still rely on it in part. The error sources separate cleanly by status:

| error source             | status                                |
|--------------------------|---------------------------------------|
| factor-node pruning      | rigorous bound ( $\delta$ sup-norm)   |
| finite envelope          | rigorous bound ( $N\Phi(-8)$ )        |
| covariance approximation | rigorous KL/Pinsker bound (Section 6) |
| $x$ -grid discretization | empirical convergence assessment      |
| factor quadrature        | empirical higher-order comparison     |
| calibration residual     | directly measured solver consistency  |

**Stability checks and two genuine bounds.** Self-convergence checks at large  $N$  show stability below what the simulation references can resolve: refining to ( $L = 3001$ , GH21) moves the shares by  $1.2 \times 10^{-8}$  at  $N = 1000$  and  $5.7 \times 10^{-8}$  at  $N = 5000$ . This is a stability statement, not by itself an error bound, since both resolutions could share bias.

Two genuine bounds come free from the product structure: conditional probabilities lie in  $[0, 1]$ , so pruning factor weight  $\delta$  perturbs unnormalized shares by at most  $\delta = 2.39 \times 10^{-8}$  in sup norm (measured:  $6.2 \times 10^{-9}$ ); and the  $\pm 8\sigma_{\max}$  envelope omits conditional mass at most  $N\Phi(-8) + \Phi(-8)^N \leq (N+1)\Phi(-8) \approx 6 \times 10^{-12}$  even at  $N = 10^4$  (the lower tail by a union bound, the upper by independence).

**Calibration accuracy at  $N = 1000$ .** At  $N = 1000$ ,  $k = 2$ , with targets from  $5 \times 10^6$  MC draws floored at  $10^{-7}$  and renormalized, the inversion completes in about three seconds. The iteration solves its own equations to  $6.4 \times 10^{-9}$ . This is a statement about the solver, not statistical accuracy, since the target carries sampling noise near  $2 \times 10^{-4}$ . It recovers the utilities of the well-resolved alternatives (target share above  $3 \times 10^{-4}$ ) to maximum error 0.017.

A replication study (experiment 16) repeats the inversion on a second  $N = 1000$  problem against three independent  $5 \times 10^6$ -draw targets: 161–162 alternatives well-resolved (98.1% of share mass), recovery error maximum 0.015–0.023 and median  $\approx 0.0027$  across replicates, with the recovery-versus-share curve reported alongside.

**Validation: where the error comes from.** Experiment 21 separates the error sources. A noiseless inverse test (targets generated by the lattice map itself, a deliberate inverse crime that isolates solver plumbing) recovers the utilities of every resolvable alternative to  $6.5 \times 10^{-7}$  at  $N = 200$  and  $1.2 \times 10^{-7}$  at  $N = 1000$  — solver error at the tolerance, so the  $\approx 0.02$  recovery in noisy tests is target sampling noise, not solver error. Alternatives with shares below the resolution floor are not chased, by design; their tier is reported separately.



**Robustness.** Twenty problems that vary utility spread, factor strength,  $D$ -heterogeneity, and rank all converged (6–31 iterations, recovery 0.004–0.024).

A further stress test targets heterogeneous idiosyncratic variances, where the common envelope is set by the largest  $\sqrt{D_i}$  while the narrowest conditional density can approach the grid spacing. At variance ratio  $10^2$  the production lattice is already converged (log-share agreement  $5 \times 10^{-12}$  against  $L = 8001$ ). At ratio  $10^3$  the production  $L = 501$  carries a maximum log-share error of  $1.0 \times 10^{-3}$ , and  $L = 2001$  restores machine precision (experiment 33). The practical rule is to scale  $L$  with the ratio of envelope width to the smallest conditional scale.

Numerical well-posedness matches Proposition 2. The reduced Jacobian assembled from  $N$  Jacobian–vector-product calls at  $N = 50$  is symmetric to  $1.7 \times 10^{-18}$ , has row-sum defect  $6.2 \times 10^{-17}$ , and reduced eigenvalues in  $[2.5 \times 10^{-5}, 0.22]$ , positive definite as proved.

**Compiled implementation.** The same pass in Rust (in the repository, with machine-precision parity and identical iteration counts) calibrates  $N = 1000$  in 3 seconds,  $N = 5000$  in 18 seconds, and  $N = 10000$  in 38 seconds (10, 11, and 13 iterations; independent  $5 \times 10^6$ -draw targets, experiment 19).

**Scaling.** Calibration is approximately linear in  $N$  over the tested range (experiment 19), a flat 11–15 forward-pass equivalents throughout, with recovery quality unchanged (median error near 0.003, 96–99% of share mass identified). At fixed spatial resolution the cost grows slowly with the utility range,  $O(\sqrt{\log N})$  under Gaussian utilities.

Figure 2 shows calibration wall time growing roughly linearly in the number of alternatives.

**Derivatives.** For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of  $\mu$  whose derivatives are evaluated without resampling. The analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. Along a utility path at  $N = 50$ , the maximum second difference over  $dt^2$  is 39.3 for fresh-draw GHK against 0.01 for both GHK under common random numbers (CRN-GHK) and the lattice. The curves are in the committed output.

These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws (Hajivassiliou et al., 1996; Hajivassiliou and McFadden, 1998; Train, 2009). The practical distinction is the character and controllability of the approximation error, quadrature resolution versus draw count, not the possibility of differentiation.

**Estimation gradients (by-product).** Experiment 30 parameterizes the covariance,  $V(\theta) = \theta_1 V_0$  and  $D(\theta) = e^{\theta_2} D_0$ , across four markets that share one utility vector and differ by known factor-scale multipliers. The outer objective is the cross-market disagreement of the calibrated utilities,  $L(\theta) = \sum_m \|\mu_m(\theta) - \bar{\mu}(\theta)\|^2$ , which vanishes at the truth. This is

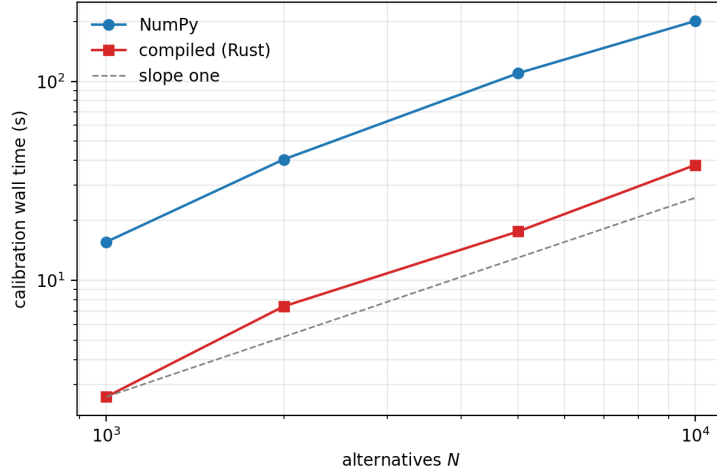


Figure 2: Wall time to calibrate the full utility vector from target shares ( $k = 2$  factors, independent  $5 \times 10^6$ -draw targets, experiment 19). Both implementations grow roughly linearly in  $N$ . The dashed guide has slope one. Recovery accuracy is assessed on alternatives above the share-resolution threshold.

a cross-market restriction of the kind that disciplines the covariance where a single menu cannot. The gradient passes through the calibration fixed point by the implicit function theorem,  $d\mu/d\theta = -B(B^\top JB)^{-1}B^\top \partial p/\partial\theta$ . In this experiment the reduced Jacobian is assembled by symmetrized central differences and solved densely at  $N = 50$ . The matrix-free route through Proposition 3 is the large- $N$  substitute. The implicit gradient matches central finite differences of the objective to  $1.2 \times 10^{-7}$ , and a damped Newton outer loop converges in three iterations to the truth up to the probit scale gauge (recovered  $\theta_1^2 e^{-\theta_2} = 1.000002$  against 1), which shares cannot identify. The scale ambiguity emerges from the experiment rather than being imposed.

**Removal ensemble (by-product).** The cached field also yields all  $N$  single-removal share vectors in  $O(QN^2L)$ , at roughly a third the wall time of recomputing them from scratch at  $N = 200$ .

Against the strongest simulation comparator, which reuses each draw’s winner and runner-up to answer every removal at once in  $O(RN + N^2)$ , the deterministic ensemble is about  $4.5\times$  more accurate at matched wall time at  $N = 200$  ( $4.2 \times 10^{-5}$  versus  $1.9 \times 10^{-4}$ ; experiment 18).<sup>9</sup>

## 5 Related work

**Prior art.** The nearest methods, and what each lacks:

<sup>9</sup>This is a side capability, not the paper’s calibration emphasis.

| approach                                                               | all $N$ shares                                            | calibration                                        |
|------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------|
| GHK simulation (Geweke et al., 1994; Hajivassiliou and McFadden, 1998) | $N$ separate runs, $R^{-1/2}$ (plain MC)                  | no reported large- $N$ share inversion known to us |
| direct utility simulation                                              | whole vector per draw, $R^{-1/2}$                         | raw frequency map is nonsmooth                     |
| analytic approximations (Bhat, 2011; Connors et al., 2014)             | per-alternative                                           | not provided                                       |
| factor quadrature (Butler and Moffitt, 1982; Elrod and Keane, 1995)    | $N$ separate $(k+1)$ -dim integrals                       | not provided                                       |
| Bayesian factor MNP (Loaiza-Maya and Nibbering, 2022)                  | simulation inside estimation                              | posterior, not a share map                         |
| neural amortization (Huch and Keane, 2026)                             | learned approximation, accuracy not resolution-controlled | not provided                                       |
| BLP-side inversion (Berry et al., 1995; Conlon and Gortmaker, 2020)    | —                                                         | logit family only                                  |
| <b>this paper</b>                                                      | one $O(QNL)$ pass                                         | Prop. 2; 3 s at $N=1000$                           |

GHK and simulated likelihood are standard in econometrics (Geweke et al., 1994; Hajivassiliou and McFadden, 1998; Train, 2009), with Hajivassiliou et al. (1996) the standard treatment of simulated MVN probabilities and their derivatives; Genz (1992) developed the same separation-of-variables construction in numerical statistics, where its RQMC form is the default MVN evaluator, anchored by the Genz–Bretz book (Genz and Bretz, 2009) and the `mvtnorm` package (Mi et al., 2009).

Deterministic grid recursions exist within that ecosystem: Miwa et al. (2003) evaluate orthant probabilities by recursion on one-dimensional lattices, at factorial cost in the dimension for general correlation, and Craig (2008) accelerates the Markov case; both price one probability per run. Ridgway (2016) reads GHK as sequential importance sampling, proves its variance can deteriorate exponentially, and builds SMC alternatives; Botev (2017) sharpened the family with minimax tilting; Genton et al. (2018) began the hierarchical line in this journal; Cao et al. (2021) push MVN probabilities to thousands of dimensions with tile-low-rank QMC, machinery pointed almost exactly at our difference covariances, which are diagonal plus rank  $k+1$ ; and Cao and Katzfuss (2025) reach linear per-probability cost past twenty thousand dimensions with Vecchia approximations.

Deterministic approximations run from Mendell–Elston (Mendell and Elston, 1974) (surprisingly competitive when optimally ordered (Connors et al., 2014)) through Bhat’s MACML (Bhat, 2011) to expectation propagation: Cunningham et al. (2011) found EP highly accurate precisely for rectangular regions, and Fasano and Denti (2025) evaluate generic Gaussian CDFs by EP on a dual probit model, deterministically and sampling-free. Every one of these prices one orthant at a time; none couples the  $N$  winner probabilities.

Factor dimension reduction for probit is old, and older than econometrics: Curnow

and Dunnett (1962) showed that product correlation collapses the multivariate normal integral to a one-dimensional integral of products of univariate CDFs, the reduction this paper conditions on, and Dunnett (1989) is its standard algorithm. The ranking-and-selection tradition built on the same integrals from the 1950s (Bechhofer, 1954; Gupta, 1963), always evaluating one designated candidate per integral: assembling all  $N$  that way costs  $N$  separate integrals where the shared field gives all  $N$  in one pass.

In econometrics, Butler and Moffitt (1982) integrated one factor by quadrature, and Lee (2000) stabilized that quadrature numerically; McFadden (1984) proposed factor-analytic MNP expressly as dimension reduction; Stern (1992) used the same conditional-independence product of normal CDFs with Monte Carlo over the residual; Bolduc (1999) handled large choice sets with parsimonious covariances but simulated probabilities. Sparse-grid integration (Heiss and Winschel, 2008) attacks the rank growth that binds product rules.

Elrod and Keane (1995) estimated factor-analytic probit for panel data; Loaiza-Maya and Nibbering (2022) pursue large choice sets by Bayesian estimation, with variational follow-ups (Loaiza-Maya and Nibbering, 2023); Kim et al. (2026) reach twenty alternatives at a million observations by neural variational inference, and Ding et al. (2024) exceed a hundred with expectation-propagation E-steps. All are approximate forward maps for estimation; none inverts shares. Girolami and Rogers (2006) use the one-dimensional product identity in its diagonal-covariance form, per class. Collier et al. (2021) place the factor-Gaussian race with low-rank-plus-diagonal covariance on the last layer of large-scale image classifiers, including thousand-class ImageNet, and approximate the winner probabilities by Monte Carlo over a temperature-softmax relaxation; Berthet et al. (2020) differentiate general perturbed-argmax layers by Monte Carlo, with  $p = \nabla G$  as the organizing identity. Both simulate the map this paper evaluates deterministically.

Racing supplied our two ingredients. Henery (1981) wrote the survival-product win integral for normal running times and gave its first-order analytic inversion. Ali (1998) inverted market odds to abilities numerically at eight-runner fields, and betting practice otherwise approximated the inversion away (Lo and Bacon-Shone, 1994). The lattice survival-product identity and its fast inverse transform come from racing (Cotton, 2021), which names the common-factor extension developed here as an open problem.

The shared-product assembly itself also appears in competing-risks computation: Mahani and Sharabiani (2019) compute all cause-specific cumulative incidences on one shared grid by forming the product of every survival function once and dividing out each cause’s own factor (software since 2015), and the Aalen–Johansen estimator (Aalen and Johansen, 1978) reads all causes off a single survival product nonparametrically. Neither embeds the assembly in factor quadrature, differentiates it, nor inverts it. Racing arrived at it independently.

Conditioning on the factors and the alternative’s own shock makes each probability a  $(k+1)$ -dimensional integral,

$$p_i = \mathbb{E}_{f,z} \prod_{j \neq i} \Phi \left( \frac{\mu_i - \mu_j + (v_i - v_j)^\top f + \sqrt{D_i} z}{\sqrt{D_j}} \right),$$

evaluable per alternative by modern randomized quasi-Monte Carlo: the natural competitor that uses exactly the structure this paper assumes. It still prices  $N$  separate integrals,  $O(RN^2)$  in total, where the shared field couples them into one  $O(QN(k + L))$  pass. Measured at matched  $4 \times 10^{-4}$  accuracy (experiment 24; the two implementations first verified against each other to  $1.7 \times 10^{-5}$ ), the per-alternative route is about  $140\times$  slower than the single shared-field pass at  $N = 1000$ , a gap that widens linearly in  $N$ .

An older transportation literature used “calibration” for MNP parameter estimation. Daganzo et al. (1977) introduced an early numerical MNP forecasting algorithm and claimed applicability to large problems, with Daganzo (1979) the monograph treatment; Sheffi et al. (1982) reviewed probability and calibration methods and proposed simulation-based estimation; Kamakura (1989) embedded the Mendell–Elston approximation in a calibration routine. These estimate parameters from choice data or approximate designated probabilities; none provides a smooth all- $N$  share map, a shared-field Jacobian–vector product, or aggregate-share inversion at the scale considered here.

Inversion theory is likewise established. Berry (1994) made share-to-utility inversion the workhorse of demand estimation; Bonnet et al. (2022) invert demand in general, including nonsmooth random-utility models, through two-sided matching; Hotz and Miller (1993) inverted choice probabilities to utility differences; Norets and Takahashi (2013) prove surjectivity of the utility-to-probability map; Berry et al. (2013) obtain demand invertibility from connected substitution; Chiong et al. (2016) invert through the convex conjugate; Fosgerau et al. (2020) show the probit inverse exists by generalized-entropy duality, without an algorithm.

Berry et al. (1995) built demand estimation on the logit-side inversion, and Conlon and Gortmaker (2020) maintain its modern practice. Chia (2021) differentiates logit BLP automatically; Huch and Keane (2026) amortize correlated-choice probabilities with neural networks, evidence of current demand for this computation.

The novelty here is accordingly not dimension reduction (Curnow and Dunnett, 1962), nor the order-statistic integral, nor the shared-product assembly in isolation (Mahani and Sharabiani, 2019; Cotton, 2021), nor well-posedness of inversion. It is the all-alternatives assembly in  $O(QNL)$ , its large- $N$  implementation, the  $O(QNL)$  Jacobian–vector product, and the shared-field reuse for calibration and counterfactuals.

## 6 Comparisons with existing algorithms

We now compare with existing approaches: first the structural arguments, then measured comparisons on shared problems, and finally the boundary regimes where the factor form itself, rather than the numerics, is the binding approximation.

**The price of smoothness.** Smoothness was never a side issue in this literature. It shaped it. The raw frequency simulator is a step function of the parameters, so Stern (1992) accepted a large per-draw cost purely to smooth it, and GHK’s adoption owed as much to producing smooth, positive, differentiable estimates (Hajivassiliou and McFadden, 1998) as to its variance reduction. The lattice ends that trade. It is smooth to

resolution, and its derivatives are exact integrals of the same pass, so the property the simulation literature paid variance and wall time for costs nothing here.

**The limits of winner-frequency simulation.** With common random numbers a smoothed simulator such as CRN-GHK is a fixed function that a solver can match to fine tolerance (a raw draw-and-argmax map is piecewise constant on the  $1/R$  grid and cannot be). The limitation is not smoothness. It is the error of that fixed function against the exact map, and the cost of driving it down.

Its coordinatewise error against the exact map is of order  $\sqrt{p_{\max}/R}$ , so making the simulated map itself  $\tau$ -accurate costs  $R \approx p_{\max}/\tau^2$  independent draws per evaluation, and inverting it exactly still leaves utilities whose true shares miss the target at a scale that is order  $\tau$  but amplified through  $J^{-1}$ .

If we take the example of direct simulation with  $N = 5000$ , this means that the forward map resolving the *exact* MNP map to  $\tau \sim 10^{-6}$  would take weeks per evaluation. A calibration could take months to a year. This comparison is at the solver’s consistency level, well beyond the  $2 \times 10^{-4}$  statistical accuracy of the noisy-target experiments.

If calibration time at a fixed accuracy can be reduced from months to a few seconds, that is a qualitative step which may materially impact the use of probit models versus their more analytically simple logit cousin. This particular example is roughly six orders of magnitude.

Common random numbers do not change the plain-MC rate. Randomized QMC and specialized Gaussian integration can improve rates as well as constants. The structural gap is different: per-alternative formulations price  $N$  separate integrals whose integrands each carry  $O(N)$  factors, and the shared field removes that factor of  $N$ . A maximum over  $N$  coordinates adds a further  $\log N$  factor in high-probability statements.

A smooth all- $N$  comparator from machine learning is the temperature-softmax relaxation  $\mathbb{E}[\text{softmax}((\mu + \xi)/T)]$  with common draws (Collier et al., 2021; Berthet et al., 2020), computing every share in  $O(RN)$ . It is biased at any fixed temperature and noisy at finite draws. Measured at  $N = 200$  against the lattice map (experiment 34), its best settings over  $T \in [0.03, 1]$  and  $R \leq 2 \times 10^5$  leave a maximum absolute share error above  $9 \times 10^{-4}$  and log-share errors above 0.1 on shares over  $10^{-3}$ , with total variation 0.43 at  $T = 1$ . Sharpening  $T$  trades bias for variance. The lattice’s corresponding log-share error is below  $5 \times 10^{-5}$ .

At the crude  $\tau = 10^{-4}$  where simulation-in-the-loop first becomes feasible, the calibrated model’s true shares miss the target at  $10^{-4}$  and the answer moves with each set of samples.

In contrast the quadrature route’s accuracy against the exact map is set by resolution, not draws, at cost independent of  $\tau$  until resolution binds. We are unaware of a reported probit share inversion at this scale in simulation-based practice, and this scaling is presumably why. We hope this work opens new territory, subject to our fixed low-factor limitation.

**General covariance through the factor approximation.** The important observation is the symmetry and arbitrary nature of absolute utilities (just as only relative horse ability matters in a race). Failing to account for this can “waste” factors.

To be more formal, the method requires  $\Sigma \approx VV^\top + \text{diag}(D)$ . GHK does not. Choices depend on  $(\mu, \Sigma)$  only through  $P\mu$  and  $P\Sigma P$ ,  $P = I - \mathbf{1}\mathbf{1}^\top/N$ , because the argmax is determined by pairwise differences. So a common factor loading  $V \rightarrow V + \mathbf{1}c^\top$  shifts every conditional location equally and cannot move the argmax (a machine-precision identity).

Our advice is that a factor fit must be run in a choice-relevant quotient space. In contrast, fitting raw  $\Sigma$  can spend scarce rank on an irrelevant common component ( $\Sigma = \tau^2\mathbf{1}\mathbf{1}^\top + \beta\beta^\top + D$  with large  $\tau$ : the raw rank-1 fit takes  $\mathbf{1}\mathbf{1}^\top$  and misses  $\beta\beta^\top$  entirely; on such an example the contrast-space fit cuts rank-1 share error from  $4.6 \times 10^{-2}$  to  $8.3 \times 10^{-3}$ ).

The asymmetry matters: only the common *factor* direction is irrelevant; a common addition to  $D$  inflates every difference variance and is choice-relevant, so  $D$  must come from the quotient fit.

The suggested fit is an iterative principal-factor heuristic applied to  $P\Sigma P$  (eigendecompose  $P\Sigma P - \text{diag}(D)$ , take the top  $k$  directions, refit  $D$  from the diagonal with a  $10^{-3}$  floor, iterate to  $10^{-10}$ ), followed by centering and SVD canonicalization of the loadings (ordered columns, fixed signs).

This procedure will not minimize a projected norm ( $P\text{diag}(D)P$  is not diagonal) however. Therefore the repository also implements an alternating least-squares fit with exact block updates on the reduced quotient (precisely, a top- $k$  PSD step for the loadings and a nonnegative least squares for  $D$  that is exact block minimization but should not be interpreted as global optimality). On the boundary matrices it changes the quotient residual by under 1%.

The covariance-quotient residual  $\|P(\Sigma - \hat{\Sigma})P\|_F / \|P\Sigma P\|_F$  is reported alongside every share error in the committed output.

We test this approach on correlation matrices with spectral decay  $\lambda_m \propto m^{-\gamma}$ ,  $\gamma \in \{0.5, 1.5, 3\}$ , at  $N = 50$  (sharing one orthogonal eigenbasis across  $\gamma$  so decay comparisons are not confounded with basis realization). The power law holds before diagonal standardization, which perturbs the spectrum; the actual leading eigenvalues (3.7, 17.1, 34.2 at  $\gamma = 0.5, 1.5, 3$ ) are printed by the script.

Share error is measured against an  $8 \times 10^6$ -draw reference, with GHK at the exact  $\Sigma$  for comparison; the plotted quantity is the rank- $k$  approximation error of the stated procedure, not a minimum over factor models.

Rank-8 errors:  $1.8 \times 10^{-3}$ ,  $3.7 \times 10^{-3}$ , and  $3.1 \times 10^{-3}$  at  $\gamma = 0.5, 1.5, 3$  from  $k = 1$  starting points of  $2.3 \times 10^{-3}$ ,  $1.5 \times 10^{-2}$ ,  $7.1 \times 10^{-2}$ .

The lattice versus simulation discrepancy is also tested. A second simulation under the fitted factor model includes the finite reference’s own noise and is flat at  $2\text{--}7 \times 10^{-4}$  across every  $(\gamma, k)$  tested. The conclusion is that the rank- $k$  error is mostly covariance-fit error, which is to say that rank is the issue, not quadrature per se.

Against GHK at  $R = 10^4$  ( $3.2, 2.3, 4.3 \times 10^{-3}$ ), rank-8 wins at  $\gamma = 0.5$  and 3 but *loses at intermediate decay*  $\gamma = 1.5$  ( $3.7 \times 10^{-3}$  versus  $2.3 \times 10^{-3}$ ). This is the hardest example. Our interpretation is that the diagonal standardization perturbs the nominal

power law. The leading eigenvalue varies from 3.7 to 34.2 across the three cases—quite different cases.

We also tried replication over three independent eigenbases. For example rank-8 beats GHK at  $R = 10^4$  in 8 of 9 basis-decay cases, with the single loss at intermediate decay. The intermediate construction seems to be getting close to the boundary where GHK should be preferred, but it is not clear cut. The wall-time is not matched. The RQMC evaluation takes about 20s versus about one second for GHK, so the latter seems to be the cheaper route to middling accuracy (the lattice’s advantage remains large  $N$  at small  $k$ ). So where accuracy demands exceed the achievable rank- $k$  error, GHK’s arbitrary- $\Sigma$  generality means that (forgive the pun) it should still be the tool of choice.

**A forward-share certificate for the covariance fit.** How much can a low-rank covariance approximation move the shares? Only the contrast part of the covariance matters, so replacing  $\Sigma$  by a fitted  $\hat{\Sigma}$  changes any share by at most

$$|p_i(\Sigma) - p_i(\hat{\Sigma})| \leq \sqrt{\frac{1}{2} \text{KL}(\mathcal{N}(B^\top \mu, B^\top \Sigma B) \parallel \mathcal{N}(B^\top \mu, B^\top \hat{\Sigma} B))}.$$

This is Pinsker’s inequality, with  $B$  any basis of  $\mathbf{1}^\perp$ . The Gaussian KL has a closed form. This turns the covariance-fit error into a guaranteed bound on the share error, computed before the transform is ever run. The bound controls the forward map at fixed  $\mu$ . It does not bound recalibrated utilities or deletion predictions, which pass through the inverse map. The bound is loose—about two orders of magnitude—but useful where it counts (experiment 26): fitting two factors to a fast-decaying four-factor truth guarantees a share error below  $5.3 \times 10^{-2}$  when the true error is  $4 \times 10^{-4}$ , and three factors guarantee  $8.4 \times 10^{-3}$  against a true  $9.7 \times 10^{-5}$ .

**Substitution: where a deleted alternative’s demand goes.** When an alternative is removed, logit must spread its demand over the survivors in proportion to their existing shares—the IIA property—while a factor model can send it preferentially to close substitutes. This experiment measures which is right, and by how much. We vary two things one at a time in a  $2 \times 2$  design, the factor structure and the idiosyncratic noise family, holding the idiosyncratic scale common so the two axes stay clean (coupling either with alternative-specific scales would confound it: an alternative-specific-scale Gumbel race is not mixed logit, and heteroskedasticity alone already breaks IIA).

Ground truth ( $N = 50$ ) is misspecified for every candidate:  $t(5)$  factors (unit variance) and skew-normal idiosyncratic noise (shape  $a = 3$ ), standardized to mean zero and unit variance, homoskedastic, with  $\mu^* \sim N(0, 1)$  and oracle loadings  $v_{ir}^* \sim N(0, 0.36/k)$ ,  $k = 2$ . (The race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates.)

The four candidates hold the idiosyncratic scale fixed and vary  $\{\text{Gumbel, Gaussian}\}$  base  $\times \{V = 0, V = V^*\}$ . The Gumbel base is variance-standardized to one, so the family axis does not alter the factor-to-idiosyncratic variance ratio.

Another comment on sign is necessary. Because the implementation is min-wins, the Gumbel base is the *reflected* (minimum) Gumbel corresponding to a standard maximum-



Gumbel in utility coordinates. The code verifies that its zero-loading case equals softmax at  $N = 12$  to  $2.8 \times 10^{-17}$ . The candidate factor law is Gaussian in all cases, so the truth’s  $t(5)$  factor law is misspecified for every candidate.

This is an *oracle-loading* design: both factor candidates receive the true  $V^*$  rather than estimating it. All four are calibrated to the same full-menu shares (residuals  $\leq 4 \times 10^{-10}$ ).

Deletion truths come from  $2 \times 10^7$  *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers.

Of 150 deletion blocks (50 singles, 100 fixed-seed random pairs), 45 with deleted mass below 0.05% are excluded as unresolvable; of the 105 scored blocks, 83 fall in the three best-resolved strata (26/29/28 at  $>10\%$  /  $2\text{--}10\%$  /  $0.5\text{--}2\%$ ) and the remaining 22, with deleted mass between 0.05% and 0.5%, appear in the final column (their truths sit closest to the common-draw resolution floor, so that column carries the widest error bars).

The score per deletion block  $B$  is the misallocated fraction of the redistributed mass,

$$\text{TV}\left(q_{\text{model}}^{(-B)}, q_{\text{truth}}^{(-B)}\right) / \sum_{i \in B} p_i,$$

averaged within strata:

|                                  | mass $> 10\%$ | 2–10%        | 0.5–2%       | 0.05–0.5%    |
|----------------------------------|---------------|--------------|--------------|--------------|
| independent Luce (= plain logit) | 0.229         | 0.245        | 0.261        | 0.305        |
| independent probit               | 0.210         | 0.232        | 0.230        | 0.267        |
| factor mixed logit               | 0.114         | 0.146        | 0.199        | 0.244        |
| factor probit                    | <b>0.045</b>  | <b>0.062</b> | <b>0.094</b> | <b>0.116</b> |

The factorial decomposition (top two strata): the factor increment is dominant in both families ( $+0.115/ +0.099$  within Gumbel,  $+0.165/ +0.170$  within Gaussian); the family increment alone is small ( $+0.019/ +0.013$  at  $V = 0$ ) but grows with factors present ( $+0.069/ +0.084$ ), a negative interaction ( $-0.050/ -0.071$ ): factor structure and the Gaussian base are complements in this design.

One caveat: the factor loadings  $v_{ir} \sim N(0, 0.36/k)$  add  $\|v_i\|^2$  to each marginal variance, unevenly across alternatives, so the factor axis also changes marginal scales—it is really a *full factor covariance* axis. The equal-marginal-variance replication (experiment 29) removes this: loadings normalized to  $\|v_i\|^2 = 0.36$ , idiosyncratic variance 0.64, so every marginal is exactly one and the axis moves only the off-diagonal covariance; twenty seeds, independent  $5 \times 10^6$ -draw targets. The pure-covariance effect is larger, not smaller: median misallocation of released demand (logit against factor probit) is 0.245 against 0.009 for deleted shares above ten percent, 0.275 against 0.031 at 2–10%, and 0.325 against 0.113 at 0.5–2%, with factor probit winning all twenty paired seeds in each of those strata; below 0.5% both degrade (0.78 against 0.68, 17 of 20) as the targets’ resolution binds.

Caveats: a single truth design and seed, and a skew-normal truth that is Gaussian-adjacent. The figure plots, for each scored single deletion, plain logit’s misallocation against factor probit’s: points below the diagonal are deletions where factor probit is closer to the truth. Singles and pairs score similarly (factor probit 0.080 versus 0.076 mean misallocated fraction) and are summarized separately in the committed output.

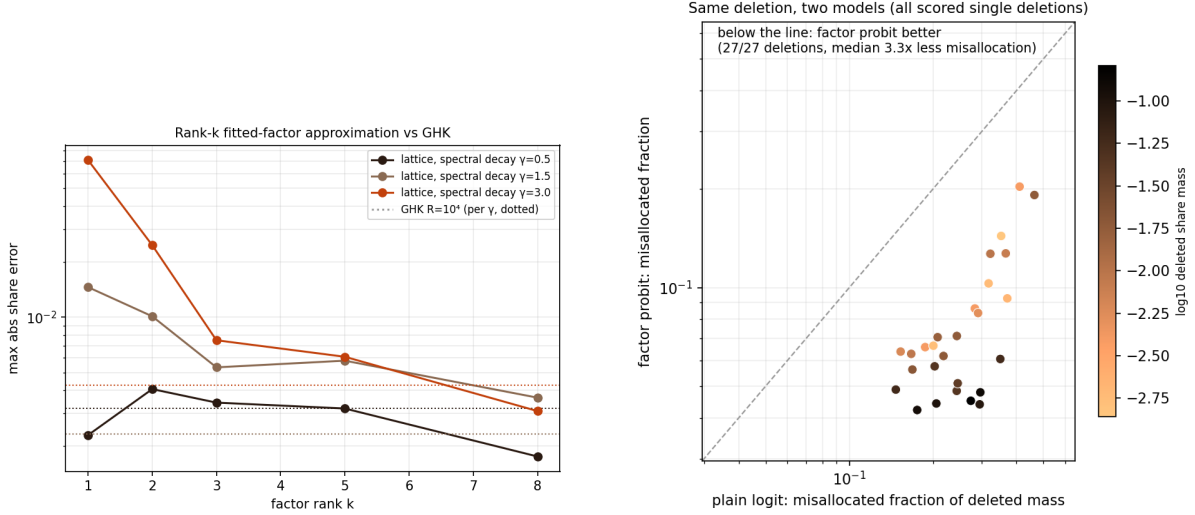


Figure 3: Left: rank- $k$  contrast-space approximation error under spectral decay  $\gamma$  (dotted: GHK at  $R = 10^4$ , per  $\gamma$ ; not wall-time matched). Right: paired comparison on identical deletions — plain logit’s misallocated fraction (horizontal) against factor probit’s (vertical), one point per scored single deletion, shaded by deleted mass; the dashed line is parity.

## 7 Discussion

Returning to our main limitation of fixed factors, estimation is the natural next step. With the reduced coordinates of Proposition 1, gradients of an outer likelihood or GMM objective pass through the calibration fixed point by implicit differentiation. So there is a route to differentiable probit demand estimation at large  $N$ .

Of course, estimating  $(V, D)$  rather than supplying them raises the usual identification cautions: scale normalization, factor rotation, and the fact that only difference covariances are identified. This is left to future work.

Separately, the pass itself can be accelerated further. Its two stages are smooth-kernel sums whose matrices are numerically low-rank in our prototype tests, and a Chebyshev-separated evaluation (in the spirit of black-box fast multipole methods) reduces each node’s cost from  $O(NL)$  to  $O(r(N + L))$ , converging quickly in  $r$  in those tests. A forward-map prototype is committed (experiment 20). Its systematic error control and extension to the derivative and calibration passes are developed in a companion paper.

Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities. A common-scale Gumbel base with zero loadings reproduces the Luce model exactly, yielding correlated-softmax generalizations.

Neither method class dominates the other. Simulation owns arbitrary  $\Sigma$  and coarse accuracy whereas the shared field we have developed owns the factor form, tight accuracy, derivatives, and calibration. Our experiments give some guidance as to the boundary.

In particular, covariance approximation rather than quadrature becomes the binding

limitation when the choice-relevant covariance is not well represented at modest rank. We leave to future work replication of the substitution factorial across truth designs and a walk-forward calibration on real assortment data.

However we hope the main message is not lost. For fifty years, computability has been a very heavy factor in deciding which choice models get used. For the factor form of probit, the form in which correlated utilities entered econometrics (Hausman and Wise, 1978), that constraint is in large part removed by our work. All shares, Jacobian–vector products, and the inverse map cost seconds at ten thousand alternatives. Integrating factor estimation with this inner solver is, again, outside the present paper, but with  $(V, D)$  in hand at modest rank, the choice between logit and factor probit becomes one of fit, not tractability.

## A Proof of Proposition 2

We show in order that the Jacobian  $J = \partial p / \partial \mu$  is the weighted graph Laplacian with weights  $w_{ij}$  (Steps 1–4), that  $J$  is positive semidefinite with null space  $\text{span}\{\mathbf{1}\}$  (Step 5), and that  $p$  restricted to  $\mathbf{1}^\perp$  is a smooth bijection onto the interior of the simplex (Steps 6–8). Throughout we work in max-wins coordinates, condition on the factor draw  $f$ , write  $m_i(f) = \mu_i + v_i^\top f$  and let  $g_i, F_i$  be the conditional Gaussian density and CDF, so that

$$p_i = \mathbb{E}_f \int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx.$$

**Step 1: off-diagonal entries.** Fix  $i \neq j$ . The mean  $\mu_j$  enters  $p_i$  only through the single factor  $F_j(x | f) = \Phi((x - m_j(f))/\sqrt{D_j})$ , whose derivative is

$$\frac{\partial}{\partial \mu_j} F_j(x | f) = -g_j(x | f).$$

Differentiation under the integral and the expectation is legitimate because the differentiated integrand is dominated, uniformly in  $(f, \mu)$ , by an integrable bound:

$$\int g_i(x | f) g_j(x | f) dx = \frac{\exp[-(m_i(f) - m_j(f))^2 / (2(D_i + D_j))]}{\sqrt{2\pi(D_i + D_j)}} \leq \frac{1}{\sqrt{2\pi(D_i + D_j)}},$$

the equality being the Gaussian product integral and the inequality bounding the exponential by one. Hence

$$J_{ij} = \frac{\partial p_i}{\partial \mu_j} = \mathbb{E}_f \int g_i \left( \frac{\partial}{\partial \mu_j} F_j \right) \prod_{\ell \neq i, j} F_\ell dx = -\mathbb{E}_f \int g_i g_j \prod_{\ell \neq i, j} F_\ell dx = -w_{ij}.$$

**Step 2: symmetry.** The integrand  $g_i g_j \prod_{\ell \neq i, j} F_\ell$  is unchanged when  $i$  and  $j$  are exchanged, so  $w_{ij} = w_{ji}$  and  $J_{ij} = J_{ji}$ .

**Step 3: diagonal entries.** Translation invariance  $p(\mu + c\mathbf{1}) = p(\mu)$  (Proposition 1) makes every row of  $J$  sum to zero,  $J\mathbf{1} = 0$ , so

$$J_{ii} = -\sum_{j \neq i} J_{ij} = \sum_{j \neq i} w_{ij}.$$

The same value follows directly. Since  $\partial_{\mu_i} g_i(x | f) = -g'_i(x | f)$ , integration by parts gives

$$\frac{\partial p_i}{\partial \mu_i} = \mathbb{E}_f \int (-g'_i) \prod_{j \neq i} F_j dx = \mathbb{E}_f \int g_i \sum_{j \neq i} g_j \prod_{\ell \neq i, j} F_\ell dx = \sum_{j \neq i} w_{ij},$$

the boundary terms vanishing because  $g_i \rightarrow 0$  at  $\pm\infty$  while  $\prod_{j \neq i} F_j$  stays bounded.

**Step 4: the Laplacian form.** Each rank-one term  $w_{ij}(e_i - e_j)(e_i - e_j)^\top$  adds  $+w_{ij}$  to the diagonal entries  $(i, i)$  and  $(j, j)$  and  $-w_{ij}$  to the off-diagonal entries  $(i, j)$  and  $(j, i)$ . Summing over unordered pairs reproduces exactly the entries of Steps 1 and 3, so

$$J = \sum_{i < j} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \quad h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2,$$

the quadratic form following from  $h^\top(e_i - e_j) = h_i - h_j$ .

**Step 5: positive definiteness on the quotient.** For every  $f$  and finite  $x$  the densities  $g_i, g_j$  are strictly positive and each  $F_\ell(x | f) \in (0, 1)$ , so the integrand of  $w_{ij}$  is strictly positive and  $w_{ij} > 0$ . The form  $h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2 \geq 0$  then vanishes only when  $h_i = h_j$  for every pair, i.e. when  $h$  is constant. Thus  $J$  is symmetric positive semidefinite with null space exactly  $\text{span}\{\mathbf{1}\}$ , and  $B^\top J B \succ 0$  for every full-rank basis  $B$  of  $\mathbf{1}^\perp$ . The identity  $p = \nabla G$  for  $G(\mu) = \mathbb{E} \max_i \{\mu_i + \xi_i\}$  is the Williams–Daly–Zachary/McFadden identity (Williams, 1977; Daly and Zachary, 1978; McFadden, 1981), so  $J = \nabla^2 G$  is the Hessian of the convex  $G$ .

**Step 6: existence of the inverse.** Fix a strictly positive target  $\hat{p}$  in the interior of the simplex and minimize

$$H(\mu) = G(\mu) - \hat{p}^\top \mu \quad \text{over } \mu \in \mathbf{1}^\perp.$$

On  $\mathbf{1}^\perp$ ,  $H$  is strictly convex, its Hessian being the reduced  $J \succ 0$  of Step 5. Since the shocks are centered, Jensen's inequality gives  $G(\mu) = \mathbb{E} \max_i (\mu_i + \xi_i) \geq \max_i (\mu_i + \mathbb{E} \xi_i) = \max_i \mu_i$ . Using  $\mathbf{1}^\top \hat{p} = 1$ ,

$$\begin{aligned} H(\mu) &\geq \max_i \mu_i - \hat{p}^\top \mu = \sum_i \hat{p}_i (\max_j \mu_j - \mu_i) \\ &\geq \hat{p}_{\min} \sum_i (\max_j \mu_j - \mu_i) \geq \hat{p}_{\min} (\max_i \mu_i - \min_i \mu_i) \\ &\geq \hat{p}_{\min} \|\mu\|_\infty \geq \hat{p}_{\min} \|\mu\|_2 / \sqrt{N}. \end{aligned}$$

The second line uses  $\hat{p}_i \geq \hat{p}_{\min} > 0$  and  $\max_j \mu_j - \mu_i \geq 0$ ; the third keeps only the single term  $i = \arg \min_i \mu_i$ ; and the fourth uses that a mean-zero vector has  $\max_i \mu_i \geq 0 \geq \min_i \mu_i$ , so its range dominates  $\|\mu\|_\infty$ . Hence  $H(\mu) \rightarrow +\infty$  as  $\|\mu\|_2 \rightarrow \infty$  on  $\mathbf{1}^\perp$ :  $H$  is coercive and attains its minimum.

**Step 7: the minimizer inverts  $p$ .** Because  $D_i > 0$ , the utilities have a smooth full-support joint density and ties are null, so  $\nabla G = p$ . At the minimizer  $\mu^*$  the gradient of  $H$  vanishes on the constraint subspace,  $B^\top(p(\mu^*) - \hat{p}) = 0$ , so  $p(\mu^*) - \hat{p} \in \text{span}\{\mathbf{1}\}$ . Both  $p(\mu^*)$  and  $\hat{p}$  sum to one, so the constant is zero and  $p(\mu^*) = \hat{p}$ . Strict convexity makes  $\mu^*$  the unique mean-zero solution.

**Step 8: smoothness.** The map  $p$  is smooth and its reduced Jacobian  $B^\top JB$  is non-singular everywhere on  $\mathbf{1}^\perp$  (Step 5), so the inverse-function theorem makes  $\hat{p} \mapsto \mu^*(\hat{p})$  smooth on the interior of the simplex. This completes the proof of Proposition 2.

## Reproducibility and Supplementary Materials

**Code and experiments:** all numbers come from committed, seeded scripts with correctness anchors run before any comparison (repository experiments 13–26, 29–31, 33, 34, and 37; 27 and 28 support the companion paper). A single manifest, `experiments/run_all_paper.py`, regenerates every table and figure; this submission is built from the immutable tag `jcgs-v1`. (Archive supplied as a zip with the submission; development repository at <https://github.com/microprediction/kinetics>.)

**Software:** the algorithm ships in an open-source Python package with an optional compiled (Rust) kernel, and package–benchmark agreement is itself a reported number.

**Data:** all experiments use synthetic data generated by the seeded scripts; no external data are required.

Hardware and software: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; compiled timings use `rustc 1.97.1` (release profile, rayon defaulting to all performance cores); experiment 17 enforces single-threaded BLAS via `threadpoolctl`; sub-second timings are warmed medians of three runs, and long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit 1.5 GB memory budget, and peak resident memory stays below 2 GB.

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## References

- H. C. W. L. Williams. On the formation of travel demand models and economic evaluation measures of user benefit. *Environment and Planning A* 9 (1977) 285–344.
- A. Daly, S. Zachary. Improved multiple choice models. In D. Hensher, M. Dalvi (eds.), *Determinants of Travel Choice*, Saxon House, 1978.
- E. P. Lazear, S. Rosen. Rank-order tournaments as optimum labor contracts. *Journal of Political Economy* 89 (1981) 841–864.
- F. Mosteller. Remarks on the method of paired comparisons I. *Psychometrika* 16 (1951) 3–9.
- C. F. Daganzo, F. Bouthelier, Y. Sheffi. Multinomial probit and qualitative choice: a computationally efficient algorithm. *Transportation Science* 11 (1977) 338–358.
- Y. Sheffi, R. Hall, C. F. Daganzo. On the estimation of the multinomial probit model. *Transportation Research Part A* 16 (1982) 447–456.
- W. A. Kamakura. The estimation of multinomial probit models: a new calibration algorithm. *Transportation Science* 23 (1989) 253–265.
- O. Bonnet, A. Galichon, Y.-W. Hsieh, K. O’Hara, M. Shum. Yogurts choose consumers? Estimation of random-utility models via two-sided matching. *Review of Economic Studies* 89 (2022) 3085–3114.
- D. McFadden. Econometric analysis of qualitative response models. In Z. Griliches, M. Intriligator (eds.), *Handbook of Econometrics*, vol. II, North-Holland, 1984, 1395–1457.
- S. Stern. A method for smoothing simulated moments of discrete probabilities in multinomial probit models. *Econometrica* 60 (1992) 943–952.
- S. Berry. Estimating discrete-choice models of product differentiation. *RAND Journal of Economics* 25 (1994) 242–262.
- D. Bolduc. A practical technique to estimate multinomial probit models in transportation. *Transportation Research B* 33 (1999) 63–79.
- J. A. Hausman, D. A. Wise. A conditional probit model for qualitative choice: discrete decisions recognizing interdependence and heterogeneous preferences. *Econometrica* 46 (1978) 403–426.
- L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34 (1927) 273–286.
- J. Geweke, M. Keane, D. Runkle. Alternative computational approaches to inference in the multinomial probit model. *Review of Economics and Statistics* 76 (1994) 609–632.

- V. Hajivassiliou, D. McFadden. The method of simulated scores for the estimation of LDV models. *Econometrica* 66 (1998) 863–896.
- K. Train. *Discrete Choice Methods with Simulation*, 2nd ed. Cambridge, 2009.
- R. J. Henery. Permutation probabilities as models for horse races. *Journal of the Royal Statistical Society B* 43 (1981) 86–91.
- M. M. Ali. Probability models on horse-race outcomes. *Journal of Applied Statistics* 25 (1998) 221–229.
- V. S. Y. Lo, J. Bacon-Shone. A comparison between two models for predicting ordering probabilities in multiple-entry competitions. *The Statistician* 43 (1994) 317–327.
- R. Loaiza-Maya, D. Nibbering. Fast variational Bayes methods for multinomial probit models. *Journal of Business & Economic Statistics* 41 (2023) 1352–1363.
- G. Kim, Y. Kang, L. Kock, P. Bansal, K. Sohn. Scalable variational inference for multinomial probit models under large choice sets and sample sizes. *Statistics and Computing* 36 (2026) 47.
- P. Ding, G. Imbens, Z. Qu, Y. Ye. Computationally efficient estimation of large probit models. arXiv:2407.09371 (2024).
- M. Collier, B. Mustafa, E. Kokiopoulou, R. Jenatton, J. Berent. Correlated input-dependent label noise in large-scale image classification. *Proceedings of CVPR* (2021) 1551–1560.
- Q. Berthet, M. Blondel, O. Teboul, M. Cuturi, J.-P. Vert, F. Bach. Learning with differentiable perturbed optimizers. *Advances in Neural Information Processing Systems* 33 (2020) 9508–9519.
- M. Girolami, S. Rogers. Variational Bayesian multinomial probit regression with Gaussian process priors. *Neural Computation* 18 (2006) 1790–1817.
- M. Fosgerau, E. Melo, A. de Palma, M. Shum. Discrete choice and rational inattention: a general equivalence result. *International Economic Review* 61 (2020) 1569–1589.
- A. S. Mahani, M. T. A. Sharabiani. Bayesian, and non-Bayesian, cause-specific competing-risk analysis for parametric and nonparametric survival functions: the R package CFC. *Journal of Statistical Software* 89 (2019) 1–29.
- O. O. Aalen, S. Johansen. An empirical transition matrix for non-homogeneous Markov chains based on censored observations. *Scandinavian Journal of Statistics* 5 (1978) 141–150.
- R. N. Curnow, C. W. Dunnett. The numerical evaluation of certain multivariate normal integrals. *Annals of Mathematical Statistics* 33 (1962) 571–579.

- C. W. Dunnett. Algorithm AS 251: multivariate normal probability integrals with product correlation structure. *Applied Statistics* 38 (1989) 564–579.
- R. E. Bechhofer. A single-sample multiple decision procedure for ranking means of normal populations with known variances. *Annals of Mathematical Statistics* 25 (1954) 16–39.
- S. S. Gupta. Probability integrals of multivariate normal and multivariate  $t$ . *Annals of Mathematical Statistics* 34 (1963) 792–828.
- J. S. Butler, R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50 (1982) 761–764.
- T. Elrod, M. P. Keane. A factor-analytic probit model for representing the market structure in panel data. *Journal of Marketing Research* 32 (1995) 1–16.
- C. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research B* 45 (2011) 923–939.
- R. Connors, S. Hess, A. Daly. Analytic approximations for computing probit choice probabilities. *Transportmetrica A* 10 (2014) 119–139.
- Z. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *JRSS-B* 79 (2017) 125–148.
- P. Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM J. Financial Mathematics* 12 (2021) 295–317.
- V. J. Hotz, R. A. Miller. Conditional choice probabilities and the estimation of dynamic models. *Review of Economic Studies* 60(3):497–529, 1993.
- S. Berry, A. Gandhi, P. Haile. Connected substitutes and invertibility of demand. *Econometrica* 81(5):2087–2111, 2013.
- S. Berry, J. Levinsohn, A. Pakes. Automobile prices in market equilibrium. *Econometrica* 63 (1995) 841–890.
- C. Conlon, J. Gortmaker. Best practices for differentiated products demand estimation with pyblp. *RAND Journal of Economics* 51 (2020) 1108–1161.
- A. Chia. Automatically differentiable random coefficient logistic demand estimation. arXiv:2106.04636 (2021).
- E. Huch, M. Keane. Amortized inference for correlated discrete choice models via equivariant neural networks. arXiv:2603.24705 (2026).
- D. McFadden. Econometric models of probabilistic choice. In C. Manski, D. McFadden (eds.), *Structural Analysis of Discrete Data*, MIT Press, 1981.



- K. Chiong, A. Galichon, M. Shum. Duality in dynamic discrete-choice models. *Quantitative Economics* 7 (2016) 83–115.
- A. Norets, S. Takahashi. On the surjectivity of the mapping between utilities and choice probabilities. *Quantitative Economics* 4 (2013) 149–155.
- R. Loaiza-Maya, D. Nibbering. Scalable Bayesian estimation in the multinomial probit model. *Journal of Business & Economic Statistics* 40 (2022) 1678–1690.
- V. Hajivassiliou, D. McFadden, P. Ruud. Simulation of multivariate normal rectangle probabilities and their derivatives: theoretical and computational results. *Journal of Econometrics* 72 (1996) 85–134.
- A. Genz, F. Bretz. *Computation of Multivariate Normal and  $t$  Probabilities*. Lecture Notes in Statistics 195, Springer, 2009.
- X. Mi, T. Miwa, T. Hothorn. mvtnorm: new numerical algorithm for multivariate normal probabilities. *The R Journal* 1 (2009) 37–39.
- T. Miwa, A. J. Hayter, S. Kuriki. The evaluation of general non-centred orthant probabilities. *Journal of the Royal Statistical Society B* 65 (2003) 223–234.
- P. Craig. A new reconstruction of multivariate normal orthant probabilities. *Journal of the Royal Statistical Society B* 70 (2008) 227–243.
- M. G. Genton, D. E. Keyes, G. Turkiyyah. Hierarchical decompositions for the computation of high-dimensional multivariate normal probabilities. *Journal of Computational and Graphical Statistics* 27 (2018) 268–277.
- J. Cao, M. Katzfuss. Linear-cost Vecchia approximation of multivariate normal probabilities. *Journal of the American Statistical Association* 121 (2026) 768–781.
- A. Genz. Numerical computation of multivariate normal probabilities. *J. Computational and Graphical Statistics* 1 (1992) 141–149.
- J. Ridgway. Computation of Gaussian orthant probabilities in high dimension. *Statistics and Computing* 26 (2016) 899–916.
- J. Cao, M. G. Genton, D. E. Keyes, G. M. Turkiyyah. Exploiting low-rank covariance structures for computing high-dimensional normal and Student- $t$  probabilities. *Statistics and Computing* 31 (2021) 2.
- N. R. Mendell, R. C. Elston. Multifactorial qualitative traits: genetic analysis and prediction of recurrence risks. *Biometrics* 30 (1974) 41–57.
- J. P. Cunningham, P. Hennig, S. Lacoste-Julien. Gaussian probabilities and expectation propagation. arXiv:1111.6832 (2011).

- A. Fasano, F. Denti. Expectation propagation for multivariate Gaussian cumulative distribution functions. *Biometrika* 112 (2025) asaf060.
- F. Aurenhammer, F. Hoffmann, B. Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica* 20 (1998) 61–76.
- B. Lévy. A numerical algorithm for  $L_2$  semi-discrete optimal transport in 3D. *ESAIM: Mathematical Modelling and Numerical Analysis* 49 (2015) 1693–1715.
- J. Kitagawa, Q. Mérigot, B. Thibert. Convergence of a Newton algorithm for semi-discrete optimal transport. *Journal of the European Mathematical Society* 21 (2019) 2603–2651.
- B. Taşkesen, S. Shafieezadeh-Abadeh, D. Kuhn. Semi-discrete optimal transport: hardness, regularization and numerical solution. *Mathematical Programming* 199 (2023) 1033–1106.
- H. Chang, Y. Narita, K. Saito. Approximating choice data by discrete choice models. Working paper, 2026 revision.
- L.-F. Lee. A numerically stable quadrature procedure for the one-factor random-component discrete choice model. *Journal of Econometrics* 95 (2000) 117–129.
- F. Heiss, V. Winschel. Likelihood approximation by numerical integration on sparse grids. *Journal of Econometrics* 144 (2008) 62–80.
- C. F. Daganzo. *Multinomial Probit: The Theory and Its Application to Demand Forecasting*. Academic Press, 1979.
- P. Cotton. The `allocation` package: Thurstone-tilt portfolio construction, 2024. <https://github.com/microprediction/allocation>.
- P. Cotton. The `winning` package: inference for multi-entrant contests, 2021. <https://github.com/microprediction/winning>.